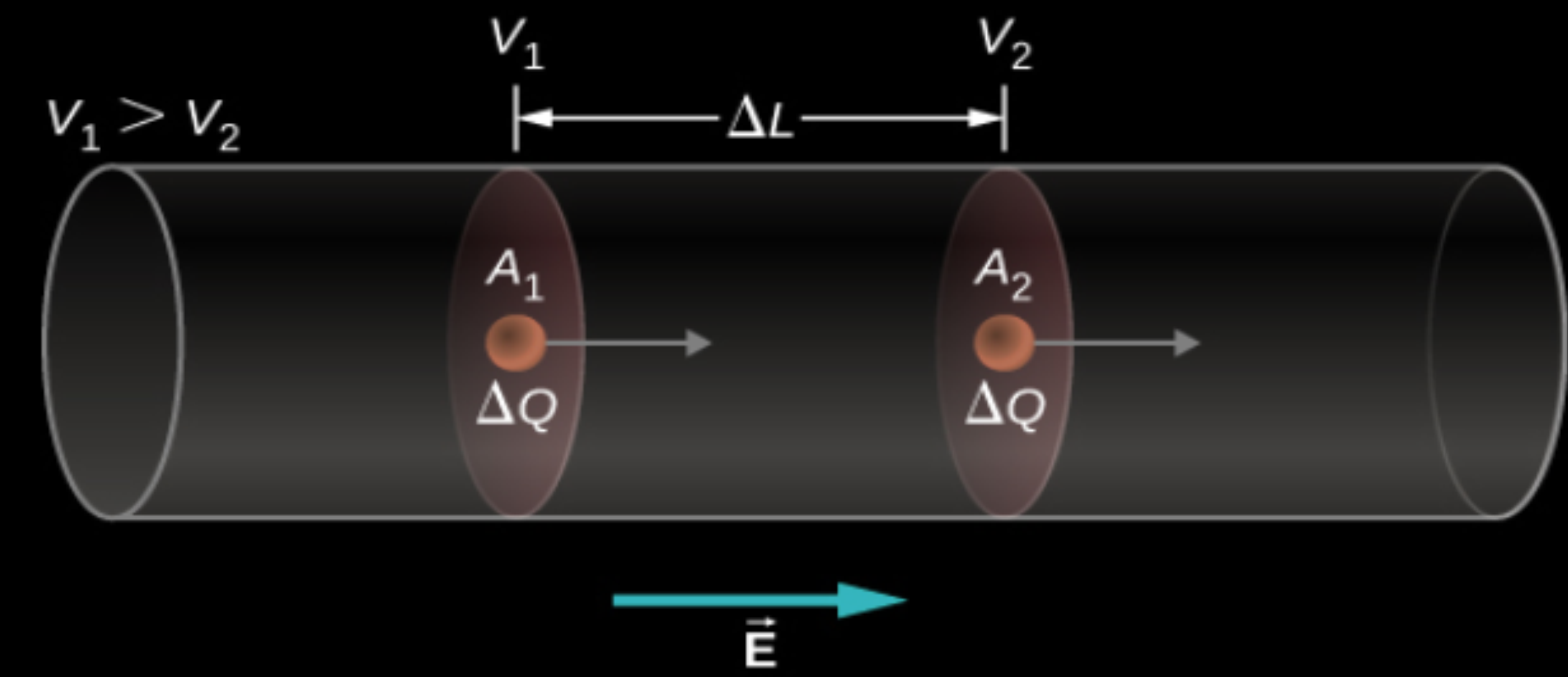


ELECTRICAL ENERGY, POWER



17. *Assertion (A) :* When three electric bulbs of power 200 W, 100 W and 50 W are connected in series to a source, the power consumed by the 50 W bulb is maximum.

Reason (R) : In a series circuit, current is the same through each bulb, but the potential difference across each bulb is different.

30. A potential difference ' V ' is applied across a load resistor of resistance R . V and R can be varied. If the current that flows in the circuit is I , draw a plot showing the variation of power consumed by the resistor as a function of :

- (a) R , keeping V constant
- (b) I , keeping R constant
- (c) V , keeping R constant

9. A conductor of $10\ \Omega$ is connected across a $6\ \text{V}$ ideal source. The power supplied by the source to the conductor is **1**
- (a) $1.8\ \text{W}$ (b) $2.4\ \text{W}$
(c) $3.6\ \text{W}$ (d) $7.2\ \text{W}$

3.	A current of 0.8 A flows in a conductor of 40 Ω for 1 minute. The heat produced in the conductor will be	1
(a)	1445 J	(b) 1536 J
(c)	1569 J	(d) 1640 J

- 25.** (a) When is more power delivered to a light bulb — just after it is turned on and the glow of the filament is increasing or after the glow becomes steady ? Why ? 2

OR

- (b) A battery is connected first across the series combination and then across the parallel combination, of three resistances R , $2R$ and $3R$. In which of the three resistances will power dissipated be maximum in the two cases ? Justify your answer. 2

1. A heater coil rated as (P, V) is cut into two equal parts. One of the parts is then connected to a battery of V volt. The power consumed by it will be

- | | |
|-------------------|-------------------|
| (A) P | (B) $\frac{P}{2}$ |
| (C) $\frac{P}{4}$ | (D) $2P$ |

20. Two electric heaters have power ratings P_1 and P_2 , at voltage V . They are connected in series to a dc source of voltage V . Find the power consumed by the combination. Will they consume the same power if connected in parallel across the same source ?

1. A wire of resistance R , connected to an ideal battery consumes a power P . If the wire is gradually stretched to double its initial length, and connected across the same battery, the power consumed will be :

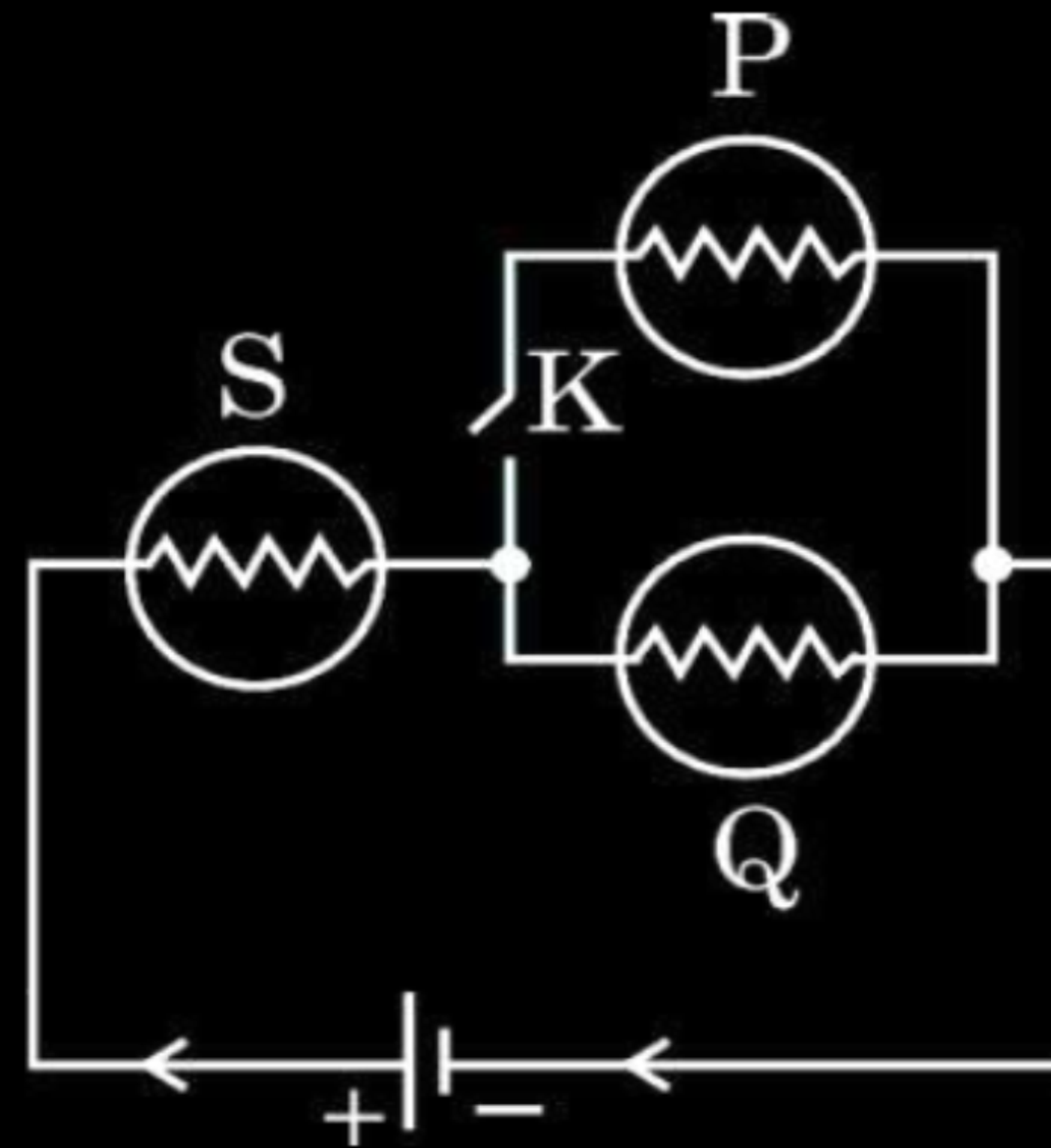
(A) $\frac{P}{4}$

(B) $\frac{P}{2}$

(C) P

(D) $2P$

- 21.** (a) In the given figure, three identical bulbs P, Q and S are connected to a battery.



- (i) Compare the brightness of bulbs P and Q with that of bulb S when key K is closed.
- (ii) Compare the brightness of the bulbs S and Q when the key K is opened.

Justify your answer in both cases.